

Reading: Focused versus Diffuse Thinking

Only dip in to read what is of interest to you - these readings are entirely optional.

Chapters 1-3 of *A Mind for Numbers* are especially helpful in providing helpful information and additional exercises related to the materials of Module 1.

Worthwhile Additional Popular Works

Access Notes: *The New York Times* has limited (5) free articles per email address. *Mind Tools* and *The New Yorker* are paid (paywall) services.

- Badre, David, "[Tips from Neuroscience to Keep you focused on hard tasks](https://doi.org/10.1038/d41586-021-00606-x)", *Nature*, 2021-03-15 doi: <https://doi.org/10.1038/d41586-021-00606-x>
- Scott Young, "[I was wrong about speed reading: Here are the facts](#)," January 2015. This excellent blog post nicely summarizes what is known in relation to speed reading.
- John Dunlosky, "[Strengthening the Student Toolbox: Study Strategies to Boost Learning](#)," *American Educator*, Fall, 2013. This excellent, comprehensive article is written by one of the top researchers in learning.
- Michael Friedman, "[Notes on Note-Taking: Review of Research and Insights for Students and Instructors](#)" has very useful insights into how to improve your note taking.
- Maria Konnikova. (January 11, 2014), "[Goodnight. Sleep Clean](#)," *The New York Times*.
- John Hamilton. (October 17, 2013). "[Brains Sweep Themselves Clean of Toxins During Sleep](#)." *NPR All Things Considered*.
- Mind Tools, "[The Pomodoro Technique® Staying Focused Throughout the Day](#)."
- Anne Trafton. (July 21, 2014), "[Try, try again? Study says no: Trying harder makes it more difficult to learn some aspects of language, neuroscientists find](#)." *Science Daily*.
- Richard C. Mohs. "[How Human Memory Works](#)." *How Stuff Works*. Notice that what Dr. Mohs calls "short term memory" in his excellent article is almost the same as

"working memory." Also, Dr. Mohs retains the "seven slots" theory of working memory--researchers still differ in their perspectives about this.

- Dan Rockmore, (Nov 7, 2019) "[The Myth and Magic of Generating New Ideas](#)," *The New Yorker*.
- Gretchen Reynolds. (April 30, 2014). "[Want to be More Creative? Take a Walk.](#)" *The New York Times*.
- Ferris Jabr, (September 3, 2014). "[Why Walking Helps Us Think.](#)" *The New Yorker*.
- Brigid Schulte, (May 16, 2014). "[For a more productive life, daydream.](#)" *CNN Opinion*.
- Robert Wright, (April 21, 2012). "[How to Break the Procrastination Habit](#)" *The Atlantic*. (Charles Duhigg's book, *The Power of Habit*, which is mentioned in the article, is also great!)
- Daniel J. Levitin, (August 9, 2014), "[Hit the Reset Button in Your Brain](#)," *The New York Times*.
- Charlie Tyson, (August 14, 2014), "[Failure to Replicate](#)," *Inside Higher Ed*. This is a very interesting overview article about the state of affairs in education research.
- Pam Harrison, (September 8, 2014), "[Sleep on It: Sleep Consolidates Memory of New Motor Task](#)," *Medscape*. Although this article deals with motor tasks, there are obvious implications related to the importance of sleep in consolidating other areas in learning. (You'll need to join to read the article, but it's free.)
- [National Numeracy](#). A website by an independent charity that is devoted to helping every person in the UK to reach a level of numeracy skills that allow them to meet their full potential.

More Recent Research

- Boyce, Richard et al. "Causal evidence for the role of REM sleep theta rhythm in contextual memory consolidation." *Science* 352, 6287 (2016): 812-816.

Heavier-Duty References (as mentioned in this week's videos)

Video: Introduction to the Focused and Diffuse Modes

- Andrews-Hanna, J.R. "The Brain's Default Network and Its Adaptive Role in Internal Mentation." *Neuroscientist* 18, no. 3 (Jun 2012): 251-70.

- Immordino-Yang, M. H., J. A. Christodoulou, and V. Singh. "Rest Is Not Idleness: Implications of the Brain's Default Mode for Human Development and Education." *Perspectives on Psychological Science* 7, no. 4 (2012): 352-64.
- Moussa MN, Steen MR, Laurienti PJ, Hayasaka S (2012) "Consistency of Network Modules in Resting-State fMRI Connectome Data." *PLoS ONE* 7(8): e44428. doi:10.1371/journal.pone.0044428.
- Raichle, Marcus E, and Abraham Z Snyder. "A Default Mode of Brain Function: A Brief History of an Evolving Idea." *NeuroImage* 37, no. 4 (2007): 1083-90.

Video: Using the Focused and Diffuse Modes: (Or, a little Dalí will do ya)

- Dali, Salvador. *Fifty Secrets of Magic Craftsmanship*. Dover, 1948 (reprint 1992).
- Root-Bernstein, Robert S., and Michelle M. Root-Bernstein. *Sparks of Genius*. NY: Houghton Mifflin, 1999.
- Takeuchi, H., Y. Taki, H. Hashizume, Y. Sassa, T. Nagase, R. Nouchi, and R. Kawashima. "The Association between Resting Functional Connectivity and Creativity." *Cerebral Cortex* 22, no. 12 (Jan 10 2012): 2921-29.
- Li, Ziyi, Ze Zhang, Tengting Tan, and Jing Luo. "[Dynamic Reconfiguration of Default and Frontoparietal Network Supports Creative Incubation](#)." *NeuroImage* (2025/01/11/ 2025): 121021. This paper notes that both focused breaks with moderate-demand cognitive activities, and light breaks with low-demand cognitive activities, can be helpful for creativity. Both strategies involve stepping away from your main task, giving your brain the space it needs to process the problem unconsciously, and returning with new insights. Choosing the type of break depends on whether you need a more creative generation or focused refinement. While reduced cognitive load promotes integration between the DMN and FPN, excessive reliance on memory replay in the hippocampus may reinforce unhelpful patterns, potentially limiting creative breakthroughs. Finding the sweet spot between mental relaxation and focus appears to be key.

- The study suggests that the two approaches differ in the type of thinking you want to enhance:

1. **Light Breaks (Low-Demand Activities):** Engage in simple, relaxing tasks like going for a walk, light exercise, or playing an easy game. These activities encourage free, associative thinking, allowing your mind to wander and form new connections. This approach is especially helpful when you're stuck and need fresh ideas.

- a. **Focused Breaks (Moderate-Demand Activities):** Perform tasks that require some concentration, like solving a puzzle, organizing something, or doing a memory-based task. These activities encourage controlled thinking and help refine or evaluate ideas you already have. This works better when you're trying to polish a solution or focus on specific details.

- Both strategies involve stepping away from your main task, giving your brain the space it needs to process the problem unconsciously, and returning with new insights. Choosing the type of break depends on whether you need a more **creative generation** or **focused refinement**.

Video: What is Learning?

- Michael D. Fox and Michael Greicius, Clinical applications of resting state functional connectivity, *Front. Syst. Neurosci.*, 16 June 2010.
- Fox, M. D., Corbetta, M., Snyder, A. Z., Vincent, J. L., and Raichle, M. E. (2006a). Spontaneous neuronal activity distinguishes human dorsal and ventral attention systems. *Proceedings of the National Academy of Sciences U.S.A.* 103, 10046–10051.
- Fox M. D., Snyder A. Z., Vincent J. L., Corbetta M., Van Essen D. C., Raichle M. E. (2005). The human brain is intrinsically organized into dynamic, anticorrelated functional networks. *Proceedings of the National Academy of Sciences U.S.A.* 102, 9673–9678. [10.1073/pnas.0504136102](https://doi.org/10.1073/pnas.0504136102).
- Guang Yang et al. Sleep promotes branch-specific formation of dendritic spines after learning, *Science* 344, 1173 (2014).

Video: A Procrastination Preview

- Boice, Robert. *Procrastination and Blocking*. Westport, CT: Praeger, 1996.
- Cremona, S., Gillig, A., Mellet, E., & Joliot, M. (2025). ["A dynamic framework of brain functional patterns shaped by spontaneous thoughts beyond the default mode network."](#) *Scientific Reports*, 15(1), 28389.
- Duan, W., Lu, P., Xu, Z., Wang, J., Lu, Y., Wang, M., Paller, K. A., Axmacher, N., & Wang, L. (2025). ["Awake reactivation of cortical memory traces predicts subsequent memory retrieval."](#) *Progress in Neurobiology*.
- Lyons, I.M., and S.L. Beilock. "When Math Hurts: Math Anxiety Predicts Pain Network Activation in Anticipation of Doing Math." *PLoS ONE* 7, no. 10 (2012): e48076.

- Steel, Piers. *The Procrastination Equation*. NY: Random House, 2010.
- Steel, Piers. "The Nature of Procrastination: A Meta-Analytic and Theoretical Review of Quintessential Self-Regulatory Failure." *Psychological Bulletin* 133, no. 1 (Jan 2007): 65-94.
- Tambini, A., & Davachi, L. (2019). "[Awake reactivation of prior experiences consolidates memories and biases cognition](#)." *Trends in Cognitive Sciences*, 23(10), 876–890.
- Tuckman, Bruce W., and Henri C. Schouwenburg. "Behavioral Interventions for Reducing Procrastination among University Students." In *Counseling the Procrastinator in Academic Settings*, edited by H.C. Schouwenburg, CH Lay, TA Pychyl and JR Ferrari Washington, DC: American Psychological Association, 2004.

Video: Practice Makes Permanent

- Brown, J.S., A. Collins, and P. Duguid. "Situated Cognition and the Culture of Learning." *Educational Researcher* 18, no. 1 (1989): 32-42.
- Dunlosky, John, Katherine A Rawson, Elizabeth J Marsh, Mitchell J Nathan, and Daniel T Willingham. "Improving Students' Learning with Effective Learning Techniques: Promising Directions from Cognitive and Educational Psychology." *Psychological Science in the Public Interest* 14, no. 1 (2013): 4-58.
- Ericsson, Karl Anders. *Development of Professional Expertise*. NY: Cambridge University Press, 2009.
- Geary, David C. *The Origin of Mind*. Washington, DC: American Psychological Association, 2005.
- Geary, David C, A Wade Boykin, Susan Embretson, Valerie Reyna, Robert Siegler, Daniel B Berch, and J Graban. "Task Group Reports of the National Mathematics Advisory Panel; Chapter 4: Report of the Task Group on Learning Processes." In, (2008) (<https://eric.ed.gov/?id=ED502980>)
- Guida, A., F. Gobet, H. Tardieu, and S. Nicolas. "How Chunks, Long-Term Working Memory and Templates Offer a Cognitive Explanation for Neuroimaging Data on Expertise Acquisition: A Two-Stage Framework." *Brain and Cognition* 79, no. 3 (Aug 2012): 221-44.
- Karpicke, Jeffrey D. "Retrieval-Based Learning Active Retrieval Promotes Meaningful Learning." *Current Directions in Psychological Science* 21, no. 3 (2012): 157-63.

- Karpicke, Jeffrey D, and Phillip J Grimaldi. "Retrieval-Based Learning: A Perspective for Enhancing Meaningful Learning." *Educational Psychology Review* 24, no. 3 (2012): 401-18.
- Karpicke, Jeffrey D, and Henry L Roediger. "The Critical Importance of Retrieval for Learning." *Science* 319, no. 5865 (2008): 966-68.
- Karpicke, Jeffrey D, Andrew C Butler, and Henry L Roediger III. "Metacognitive Strategies in Student Learning: Do Students Practice Retrieval When They Study on Their Own?". *Memory* 17, no. 4 (2009): 471-79.
- Karpicke, J. D., and J. R. Blunt. "Retrieval Practice Produces More Learning Than Elaborative Studying with Concept Mapping." *Science* 331, no. 6018 (Feb 11 2011): 772-5.
- Karpicke, J.D., and J.R. Blunt. "Response to Comment on 'Retrieval Practice Produces More Learning Than Elaborative Studying with Concept Mapping'." *Science* 334, no. 6055 (2011): 453-53.
- Kornell, Nate, Matthew Jensen Hays, and Robert A Bjork. "Unsuccessful Retrieval Attempts Enhance Subsequent Learning." *Journal of Experimental Psychology: Learning, Memory, and Cognition* 35, no. 4 (2009): 989.
- Kornell, N., A. D. Castel, T. S. Eich, and R. A. Bjork. "Spacing as the Friend of Both Memory and Induction in Young and Older Adults." *Psychology and Aging* 25, no. 2 (Jun 2010): 498-503.
- McDaniel, M. A., and A. A. Callender. "Cognition, Memory, and Education." In *Cognitive Psychology of Memory, Vol 2 of Learning and Memory*, edited by Henry L Roediger. 819-43. Oxford, UK: Elsevier, 2008.
- Roediger, Henry L., and Mary A. Pyc. "Inexpensive Techniques to Improve Education: Applying Cognitive Psychology to Enhance Educational Practice." *Journal of Applied Research in Memory and Cognition* 1, no. 4 (2012): 242-48.
- Roediger, Henry L, and Andrew C Butler. "The Critical Role of Retrieval Practice in Long-Term Retention." *Trends in Cognitive Sciences* 15, no. 1 (2011): 20-27.
- Roediger, Henry L, and Jeffrey D Karpicke. "The Power of Testing Memory: Basic Research and Implications for Educational Practice." *Perspectives on Psychological Science* 1, no. 3 (2006): 181-210.
- Rohrer, Doug, and Harold Pashler. "Increasing Retention without Increasing Study Time." *Current Directions in Psychological Science* 16, no. 4 (2007): 183-86.

- Taylor, Kelli, and Doug Rohrer. "The Effects of Interleaved Practice." *Applied Cognitive Psychology* 24, no. 6 (2010): 837-48.
- See also extensive endnote references and discussions in Chapters 2 and 3, *A Mind for Numbers*, Barbara Oakley, Penguin, 2014.

Video: Introduction to Memory

- Baddeley, A., Eysenck, M. W., & Anderson, M. C. (2009). *Memory*. NY: Psychology Press.
- Carpenter, S. K., Cepeda, N. J., Rohrer, D., Kang, S. H. K., & Pashler, H. (2012). Using spacing to enhance diverse forms of learning: Review of recent research and implications for instruction. *Educational Psychology Review*, 24(3), 369-378. doi: 10.1007/s10648-012-9205-z
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- Dudai, Y. (2004). The neurobiology of consolidations, or, how stable is the engram? *Annual Review of Psychology*, 55, 51-86.
- Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques: Promising directions from cognitive and educational psychology. *Psychological Science in the Public Interest*, 14(1), 4-58.
- Guida, A., Gobet, F., Tardieu, H., & Nicolas, S. (2012). How chunks, long-term working memory and templates offer a cognitive explanation for neuroimaging data on expertise acquisition: A two-stage framework. *Brain and Cognition*, 79(3), 221-244. doi: 10.1016/j.bandc.2012.01.010
- Rawson, K. A., & Dunlosky, J. (2011). Optimizing schedules of retrieval practice for durable and efficient learning: How much is enough? *Journal of Experimental Psychology: General*, 140(3), 283.
- Rohrer, Doug, Robert F. Dedrick, and Kaleena Burgess. "The Benefit of Interleaved Mathematics Practice Is Not Limited to Superficially Similar Kinds of Problems." *Psychonomic Bulletin & Review* in press (2013).
- Rohrer, D., & Pashler, H. (2010). Recent research on human learning challenges conventional instructional strategies. *Educational Researcher*, 39(5), 406-412.

Video: The Importance of Sleep in Learning

- Djonlagic, I., A. Rosenfeld, D. Shohamy, C. Myers, M. Gluck, and R. Stickgold. "Sleep Enhances Category Learning." *Learning & Memory* 16, no. 12 (Dec 2009): 751-5.
- Durrant, Simon J., Scott A. Cairney, and Penelope A. Lewis. "Overnight consolidation aids the transfer of statistical knowledge from the medial temporal lobe to the striatum." *Cerebral Cortex* 23.10 (2013): 2467-2478.
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- Ellenbogen, J.M., P.T. Hu, J.D. Payne, D. Titone, and M.P. Walker. "Human Relational Memory Requires Time and Sleep." *PNAS*, 104, no. 18 (2007): 7723-28.
- Erlacher, Daniel, and Michael Schredl. "Practicing a Motor Task in a Lucid Dream Enhances Subsequent Performance: A Pilot Study." *The Sport Psychologist*, 24, no. 2 (2010): 157-67.
- Moss, R. *The Secret History of Dreaming*. Novato, CA: New World Library, 2008.
- Scullin, M. K., and M. A. McDaniel. "Remembering to Execute a Goal: Sleep on It!" *Psychological Science* 21, no. 7 (Jul 2010): 1028-35.
- Stickgold, Robert, and Jeffrey M Ellenbogen. "Quiet! Sleeping Brain at Work." *Scientific American Mind* 19, no. 4 (2008): 22-29.
- Wamsley, Erin J., Matthew Tucker, Jessica D. Payne, Joseph A. Benavides, and Robert Stickgold. "Dreaming of a Learning Task Is Associated with Enhanced Sleep-Dependent Memory Consolidation." *Current Biology*, 20, no. 9 (2010): 850-55.
- Xie, Lulu, Hongyi Kang, Qiwu Xu, Michael J Chen, Yonghong Liao, Meenakshisundaram Thiyagarajan, John O'Donnell, et al. "Sleep Drives Metabolite Clearance from the Adult Brain." *Science*, 342, no. 6156 (2013): 373-77.

In Video Questions

[Here's](#) a pdf of the questions and answers found in the videos.